RAJALAKSHMI ENGINEERING COLLEGE

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M2

Started on	Tuesday, 25 August 2026, 10:58 AM
State	Finished
Completed on	Tuesday, 25 August 2026, 1:51 PM
Time taken	2 hours 53 mins
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
    {
         i++;
         s += i;
     }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

void function(int n) {
    int count = 0;


    int i = 1; count++;
    int s = 1; count++;


    count++;
    while(s <= n) {
        i++;    count++;
        s += i; count++;

        count++;
    }


    printf("%d\n", count);
}

int main() {
    int n;
    if (scanf("%d", &n) == 1) {
        function(n);
    }
    return 0;
}
```

	Input	Expected	Got	
✔	9	12	12	✔

	Input	Expected	Got	
✔	4	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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